

Aws-Security course

AWS Security Course

Final Project – Flaws2.Cloud Security Labs

Objective 1 – Connecting to the Security Account via AWS CLI

The first objective of this project was to establish a secure connection to the **Security** AWS account using the **AWS Command Line Interface (CLI)** and download the organization's **AWS CloudTrail** logs for further security analysis.

Authentication was successfully completed using an **AWS Access Key ID** and **AWS Secret Access Key**.

The AWS CLI was configured using the following command:

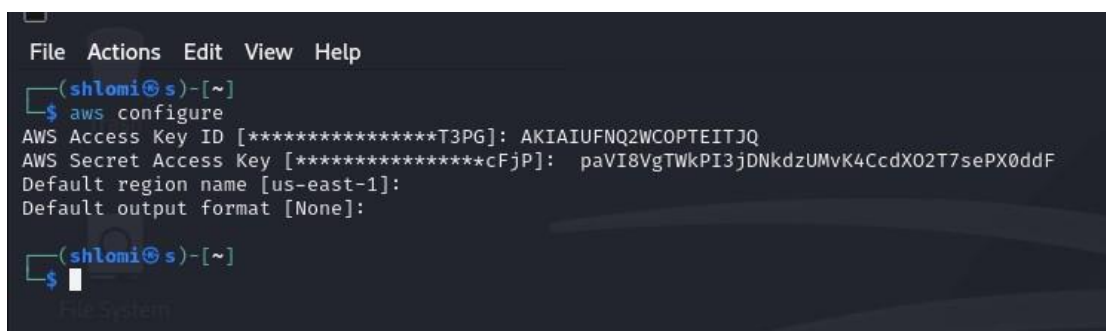
```
```bash
aws configure
```
```

During the configuration process, the following parameters were provided:

- AWS Access Key ID
- AWS Secret Access Key
- Default Region (`us-east-1`)
- Default Output Format

The **us-east-1** region was selected because the CloudTrail log files used throughout the lab were stored in an Amazon S3 bucket located in that region.

The screenshot above demonstrates the successful configuration of the AWS CLI using the Security account credentials.



```
File Actions Edit View Help
(shlomi@shlomi)-[~]
$ aws configure
AWS Access Key ID [*****T3PG]: AKIAIUFNQ2WCOPTEITJQ
AWS Secret Access Key [*****cFjP]: 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
Default region name [us-east-1]:
Default output format [None]:

(shlomi@shlomi)-[~]
$
```

Connection Verification

To verify that the AWS CLI was successfully authenticated to the **Security** account, the following command was executed:

```
```bash
aws sts get-caller-identity
```
```

This command queries the **AWS Security Token Service (STS)** and returns information about the identity currently associated with the active CLI session, including the AWS Account ID, User ARN, and User ID.

The successful output confirmed that the AWS CLI was correctly authenticated and connected to the intended Security account before proceeding with the CloudTrail log collection. `Aws sts get-caller-identity`

```
File Actions Edit View Help
[shlomi@~]$ aws configure
AWS Access Key ID [*****ITJQ]:
AWS Secret Access Key [*****0ddF]:
Default region name [us-east-1]:
Default output format [None]:

[shlomi@~]$ aws sts get-caller-identity
{
  "UserId": "AIDA3XZBU42TNRNGBBFI",
  "Account": "322079859186",
  "Arn": "arn:aws:iam::322079859186:user/security"
}

[shlomi@~]$ aws s3 sync s3://Flaws2-logs /home/shlomi/sec_course
download: s3://Flaws2-logs/AWSLogs/653711331788/CloudTrail/us-east-1/2018/11/28/653711331788_CloudTrail_us-east-1_20181128T2310Z_rp9i9zxR2Vcpqfnz.json.gz to AWSLogs/653711331788/CloudTrail/us-east-1/2018/11/28/653711331788_CloudTrail_u
s-east-1_20181128T2310Z_rp9i9zxR2Vcpqfnz.json.gz
download: s3://Flaws2-logs/AWSLogs/653711331788/CloudTrail/us-east-1/2018/11/28/653711331788_CloudTrail_us-east-1_20181128T2352_cr9ra70HlrytWYXV.json.gz to AWSLogs/653711331788/CloudTrail/us-east-1/2018/11/28/653711331788_CloudTrail_u
s-east-1_20181128T2352_cr9ra70HlrytWYXV.json.gz
download: s3://Flaws2-logs/AWSLogs/653711331788/CloudTrail/us-east-1/2018/11/28/653711331788_CloudTrail_us-east-1_20181128T23052_83VTWZ820kiEC7Lq.json.gz to AWSLogs/653711331788/CloudTrail/us-east-1/2018/11/28/653711331788_CloudTrail_u
s-east-1_20181128T23052_83VTWZ820kiEC7Lq.json.gz
download: s3://Flaws2-logs/AWSLogs/653711331788/CloudTrail/us-east-1/2018/11/28/653711331788_CloudTrail_us-east-1_20181128T2310Z_jQa3Cuioboj08I4y.json.gz to AWSLogs/653711331788/CloudTrail/us-east-1/2018/11/28/653711331788_CloudTrail_u
s-east-1_20181128T2310Z_jQa3Cuioboj08I4y.json.gz
download: s3://Flaws2-logs/AWSLogs/653711331788/CloudTrail/us-east-1/2018/11/28/653711331788_CloudTrail_us-east-1_20181128T23052_zKlMhON7EphLa9u.json.gz to AWSLogs/653711331788/CloudTrail/us-east-1/2018/11/28/653711331788_CloudTrail_u
s-east-1_20181128T23052_zKlMhON7EphLa9u.json.gz
download: s3://Flaws2-logs/AWSLogs/653711331788/CloudTrail/us-east-1/2018/11/28/653711331788_CloudTrail_us-east-1_20181128T2310Z_739NEixrjJsrLXsd.json.gz to AWSLogs/653711331788/CloudTrail/us-east-1/2018/11/28/653711331788_CloudTrail_u
s-east-1_20181128T2310Z_739NEixrjJsrLXsd.json.gz
download: s3://Flaws2-logs/AWSLogs/653711331788/CloudTrail/us-east-1/2018/11/28/653711331788_CloudTrail_us-east-1_20181128T2310Z_j3W5HfNtz7K0nvcP.json.gz to AWSLogs/653711331788/CloudTrail/us-east-1/2018/11/28/653711331788_CloudTrail_u
s-east-1_20181128T2310Z_j3W5HfNtz7K0nvcP.json.gz
download: s3://Flaws2-logs/AWSLogs/653711331788/CloudTrail/us-east-1/2018/11/28/653711331788_CloudTrail_us-east-1_20181128T2310Z_A1lhv3sWzR1BfVkv.json.gz to AWSLogs/653711331788/CloudTrail/us-east-1/2018/11/28/653711331788_CloudTrail_u
s-east-1_20181128T2310Z_A1lhv3sWzR1BfVkv.json.gz

[shlomi@~]$
```

AWS Security Token Service (STS)

AWS Security Token Service (STS) is a global AWS service that provides temporary security credentials for authenticated users and services.

Instead of relying solely on long-term credentials, STS generates temporary **Access Key ID**, **Secret Access Key**, and **Session Token** credentials with a

configurable expiration time. This approach reduces the risk associated with long-term credential exposure and is considered an AWS security best practice.

When creating an AWS account or assuming an IAM role, AWS often uses STS behind the scenes to issue temporary credentials, even though this process is typically transparent to the user.

Verifying the Authenticated Identity

The following command was executed to verify the identity associated with the active AWS CLI session:

```
```bash
aws sts get-caller-identity
```
```

The `get-caller-identity` API returns information about the currently authenticated IAM identity, including:

- AWS Account ID
- User ID
- Amazon Resource Name (ARN)

The successful response confirmed that the AWS CLI session was authenticated using the intended IAM identity and that communication with the Security account had been successfully established.

Downloading CloudTrail Logs

The next step consisted of downloading the organization's **AWS CloudTrail** logs for further security analysis.

CloudTrail is a native AWS logging service that records management events and API activity across all AWS accounts within an AWS Organization. Every AWS API request—including the identity that initiated the request, the executed action, the source IP address, and the event timestamp—is recorded and can later be used for security investigations, auditing, and incident response.

CloudTrail logs can be integrated with **Amazon CloudWatch** for monitoring and alerting or stored in **Amazon S3** for long-term retention and forensic analysis.

The logs were downloaded using the following command:

```
```bash
aws s3 sync s3://flaws2-logs .
```
```

Command Explanation

``sync``

Synchronizes the contents of the source and destination directories by copying only new or modified files. Compared to the ``cp`` command, ``sync`` is more efficient when working with large datasets because it avoids downloading files that already exist locally.

`**`s3://flaws2-logs`**`

Specifies the Amazon S3 bucket containing the CloudTrail log files collected from the Security account.

The downloaded log files were successfully synchronized to the local Kali Linux workstation, providing a complete dataset for subsequent CloudTrail analysis and security investigations.

Objective 2 – Accessing the Target Account

AWS recommends following the **multi-account security model**, where security services, audit logs, and monitoring capabilities are centralized in a dedicated **Security** account rather than being stored within individual workload accounts.

Following this security best practice, the CloudTrail logs used throughout this project were stored in a separate **Target** AWS account. To access these logs securely, a cross-account trust relationship was established between the **Security** account and the **Target** account.

This architecture provides several security advantages:

- Centralized storage of audit logs.
- Protection against log tampering by compromised workload accounts.
- Separation of duties between production and security environments.
- Simplified incident response and forensic investigations.

To analyze the CloudTrail logs, the Security account was configured to securely assume an IAM role in the Target account, granting temporary access to the required resources without exposing long-term credentials.

This implementation follows the AWS **Security Best Practices** for cross-account access and centralized logging within multi-account environments.

```
(shlomi@shlomi)-[~/aws]
$ cat config
[profile security]
region = us-east-1
output = json
[profile shlomi]
region = us-east-1
[profile ifsafeuser]
region = us-east-1
output = json
[profile itsafeuser]
region = us-east-1d
[profile target_security]
region=us-east-1
output=json
source_profile = security
role_arn = arn:aws:iam::653711331788:role/security

(shlomi@shlomi)-[~/aws]
$ aws --profile target_security sts get-caller-identity
{
  "UserId": "AROAIIKRY5GULQLYOGRMNS:botocore-session-1744830276",
  "Account": "653711331788",
  "Arn": "arn:aws:sts::653711331788:assumed-role/security/botocore-session-1744830276"
}

(shlomi@shlomi)-[~/aws]
$ aws --profile target_security s3 ls
2018-11-20 21:50:08 flaws2.cloud
2018-11-20 20:45:26 level1.flaws2.cloud
2018-11-21 03:41:16 level2-g9785tw8478k4awxtbox9kk3c5ka8iiz.flaws2.cloud
2018-11-26 21:47:22 level3-oc6ou6dnkw8sszwvdrxaxc5t5udrsw3s.flaws2.cloud
2018-11-27 22:37:27 the-end-962b72bjahfm5b4wcktm8t9z4sapemjb.flaws2.cloud

(shlomi@shlomi)-[~/aws]
$
```

Configuring Cross-Account Access

After the initial authentication to the **Security** account using the AWS CLI, a hidden directory named **aws** was automatically created in the user's home directory

The directory can be displayed using the following command

```
bash``
ls -a
``
```

The **aws** directory contains two important configuration files

config – Stores AWS CLI profiles, regions, output formats, and role-assumption settings -
credentials – Stores the Access Key ID and Secret Access Key associated with authenticated AWS -
accounts

To enable secure access to the **Target** account, a new AWS CLI profile was added to the **config** file

The profile was configured to

- .Use the **Security** account as the source profile -
- .Assume the IAM role provided by the Target account (`role_arn`) -
- .**Create a dedicated CLI profile named target_security** -

This configuration enables secure cross-account authentication without requiring additional long-term credentials

Verifying Cross-Account Authentication

:After configuring the new profile, the following command was executed

```
bash``
aws --profile target_security sts get-caller-identity
``
```

Command Explanation

- .profile target_security - Executes the command using the Target account profile-- -
- .sts - Invokes the AWS Security Token Service to obtain temporary credentials -
- get-caller-identity - Returns information about the IAM identity currently associated with the -
- .active session

The output confirmed that the CLI had successfully assumed the IAM role in the **Target** account.
The returned ARN identified the session as an **AssumedRole**, verifying that cross-account authentication was functioning correctly

Listing the CloudTrail Storage Bucket

Once authentication to the Target account was confirmed, the contents of the Amazon S3 bucket were listed using the following command

```
bash``
aws --profile target_security s3 ls
``
```

Command Explanation

- .profile target_security - Executes the command using the Target account profile-- -
- .s3 - Invokes the Amazon S3 service -
- .ls - Lists the available buckets or objects within a bucket -

The successful output displayed the CloudTrail log storage location, confirming that the assumed IAM role had the required permissions to access the CloudTrail data stored in the Target account

This step verified that the cross-account trust relationship had been correctly configured and that the Security account could securely access audit logs located in the Target account without exposing permanent credentials

Objective 3 – CloudTrail Log Analysis Using jq

##The third objective focused on processing and analyzing AWS CloudTrail log files using jq

jq is a command-line utility used to parse, filter, transform, and format JSON data. Because CloudTrail logs are stored in JSON format, jq makes it possible to extract specific fields and convert large log datasets into structured, readable output suitable for investigation

Decompressing the CloudTrail Log Files

CloudTrail log files are commonly stored in compressed .gz format. Before analyzing them, the files were recursively located and decompressed using the following command

```
bash``
;\ {} find . -type f -exec gunzip
``
```

Command Explanation

.find`\*\*` – Searches for files and directories based on defined criteria`\*\*` -
Starts the search from the current directory – `\*\*`.`\*\*` -
.type f`\*\*` – Restricts the results to regular files-`\*\*` -
.`exec`**` – Executes another command against each file returned by `find`-`**` -
.gunzip`\*\*` – Decompresses files stored in gzip format`\*\*` -
Represents the current file found by the `find` command – `\*\*` `{}` `\*\*` -
.Terminates the `-exec` command – `\*\*`,`\*\*` -

This command recursively searched the current CloudTrail directory and decompressed each matching .log file

Formatting JSON Logs with jq

:After decompressing the logs, the contents of all files were passed to jq using the following command

```
bash``
'| find . -type f -exec cat {} \; |jq
``
```

Command Explanation

.cat`\*\* – Reads and outputs the contents of each file`** -
 .Passes the output of the previous command to the next command – **`|`** -
.jq `'|`** – Parses and formats the entire JSON document without applying a filter`** -

The dot filter instructs jq to return the complete JSON structure. This produces a cleaner and more .readable output, although the result still contains a large amount of information

Extracting CloudTrail Event Names

To reduce the amount of displayed data, the next command extracted only the event names recorded in :the CloudTrail logs

```
bash``
|;\ {} find . -type f -exec cat
```

```
'jq -r '.Records[] | .eventName
```

```
'''
```

Command Explanation

`.Records[]`**` – Iterates through every event stored in the CloudTrail `'Records'` array.`** -

`.eventName`**` – Extracts the AWS API operation recorded for each event.`** -

`.r`**` – Outputs the values as raw text without JSON quotation marks-`** -

This output displayed the actions performed in the AWS account, such as authentication activity, resource access, policy changes, and API requests

.However, event names alone were not sufficient for building a complete activity timeline

```
---
```

Creating a Chronological Event Timeline

:To add the timestamp associated with each AWS event, the following command was executed

```
bash``
```

```
|;\ {} find . -type f -exec cat
```

```
| 'jq -r '.Records[] | [.eventTime, .eventName] | @tsv
```

```
sort
```

```
'''
```

Command Explanation

`.eventTime`**` – Extracts the timestamp of the event.`** -

`.eventName`**` – Extracts the AWS API action.`** -

`.Combines the selected fields into an array – **'[ ... ]`** -`

`.tsv`**` – Converts the array into tab-separated values@`** -

.sort`\*\* – Sorts the resulting output in ascending order`** -

The resulting output created a simple timeline containing the time and name of each recorded AWS
.event

Enriching the CloudTrail Dataset

:To support deeper security analysis, additional identity and network fields were added to the jq query

```
bash``
|;\ {} find . -type f -exec cat
'jq -r
| []Records.
]
,eventTime.
,sourceIPAddress.
,userIdentity.arn.
,userIdentity.accountId.
,userIdentity.type.
eventName.
|[
tsv@
|'
sort
``
```

Extracted Fields

.eventTime`\*\* – The date and time when the event occurred`** -

.sourceIPAddress`\*\* – The source IP address from which the AWS API request originated`** -

.userIdentity.arn`\*\* – The Amazon Resource Name of the identity associated with the event`** -

.userIdentity.accountId`\*\* – The AWS account ID associated with the identity`** -

userIdentity.type`\*\* – The type of identity that performed the action, such as an IAM user, assumed `** -

.role, AWS service, or root account

.eventName`\*\* – The AWS API operation that was performed`** -

:This enriched output provided a structured view of

.When the event occurred -

.Where the request originated -

.Which identity performed the action -

.Which AWS account was involved -

.What type of identity was used -

.Which API operation was executed -

Exporting the Analysis Results

:The final output was redirected into a local report file for easier review and further analysis

```
bash``
|;\ {} find . -type f -exec cat
'jq -r
| []Records.
]
,eventTime.
,sourceIPAddress.
,userIdentity.arn.
,userIdentity.accountId.
,userIdentity.type.
eventName.
```

```
| [  
tsv@  
| '  
sort > report.txt  
'''
```

:The `>` operator redirects the standard output of the command into a file named

```
text'''  
report.txt  
'''
```

The generated report could then be reviewed locally on the Kali Linux workstation or imported into tools
.such as Microsoft Excel for filtering, sorting, and further investigation

Security Benefit

Using jq to analyze CloudTrail logs makes it possible to quickly transform large volumes of JSON data
.into a structured investigation dataset

:This approach supports

- .AWS activity timeline reconstruction -
- .Identification of suspicious API operations -
- .Analysis of source IP addresses -
- .Tracking of IAM users and assumed roles -
- .Detection of unauthorized account activity -
- .Export of cloud audit data into external investigation tools -

The final report provided a concise and searchable view of the activity that occurred within the AWS .environment

```
shlomi@shlomi:~/aws_course/AWS_logs/653711331788/CloudTrail/us-east-1/2018/11/28
$ ls
653711331788_CloudTrail_us-east-1_20181128T22355Z_c89u70HryWkvY.json.gz  653711331788_CloudTrail_us-east-1_20181128T2300Z_79M6TcrJho7KMD.json.gz  653711331788_CloudTrail_us-east-1_20181128T2310Z_j0aCuihoj08Tay.json.gz
653711331788_CloudTrail_us-east-1_20181128T2305Z_83V7M2G2M4iC7L4.json.gz  653711331788_CloudTrail_us-east-1_20181128T2310Z_A1lhz24mc828FM.json.gz  653711331788_CloudTrail_us-east-1_20181128T2310Z_r9b0ea820ipufm.json.gz
653711331788_CloudTrail_us-east-1_20181128T2305Z_r6Lm0M7Egula9u.json.gz  653711331788_CloudTrail_us-east-1_20181128T2310Z_79M6TcrJho7KMD.json.gz

(shlomi@shlomi)-[~/../us-east-1/2018/11/28]
$ find . -type f -exec cat {} \;

(shlomi@shlomi)-[~/../us-east-1/2018/11/28]
$ ls
653711331788_CloudTrail_us-east-1_20181128T22355Z_c89u70HryWkvY.json  653711331788_CloudTrail_us-east-1_20181128T2300Z_79M6TcrJho7KMD.json  653711331788_CloudTrail_us-east-1_20181128T2310Z_j0aCuihoj08Tay.json
653711331788_CloudTrail_us-east-1_20181128T2305Z_83V7M2G2M4iC7L4.json  653711331788_CloudTrail_us-east-1_20181128T2310Z_A1lhz24mc828FM.json  653711331788_CloudTrail_us-east-1_20181128T2310Z_r9b0ea820ipufm.json
653711331788_CloudTrail_us-east-1_20181128T2305Z_r6Lm0M7Egula9u.json  653711331788_CloudTrail_us-east-1_20181128T2310Z_79M6TcrJho7KMD.json

(shlomi@shlomi)-[~/../us-east-1/2018/11/28]
$ find . -type f -exec cat {} \; | jq '.Records[]'

{
  "Records": [
    {
      "eventVersion": "1.05",
      "userIdentity": {
        "type": "AssumedRole",
        "principalId": "AROA3QWBNWJML2XNF64:d190d1a-248e-45d6-9113-46da22d7f2c7",
        "arn": "arn:aws:iam::653711331788:assumed-role/level3/d190d1a-248e-45d6-9113-46da22d7f2c7",
        "accountId": "653711331788",
        "accessKeyId": "ASIA2QNB3HGHGKWB5J5",
        "sessionContext": {
          "attributes": {
            "MFAAuthenticated": "false",
            "creationDate": "2018-11-28T22:31:59Z"
          },
          "sessionIssuer": {
            "type": "Role",
            "principalId": "AROA3QWBNWJML2XNF64",
            "arn": "arn:aws:iam::653711331788:role/level3",
            "accountId": "653711331788",
            "userName": "level3"
          }
        }
      },
      "eventName": "AssumeRole",
      "eventTime": "2018-11-28T23:09:28Z",
      "eventSource": "s3.amazonaws.com",
      "eventDetail": "ListBuckets",
      "awsRegion": "us-east-1",
      "sourceIPAddress": "206.102.221.258",
      "userAgent": "aws-cli/1.16.19 Python/2.7.10 Darwin/17.7.0 boto3/1.12.91",
      "requestParameters": null,
      "responseElements": null,
      "requestID": "4698593893188277",
      "eventID": "65c1118-834e-4d48-9073-1021a480473"
    }
  ]
}
```

```
(shlomi@shlomi)-[~/../us-east-1/2018/11/28]
$ find . -type f -exec cat {} \; | jq '.Records[]|eventName'

"ListBuckets"
"GetObject"
"GetObject"
"GetObject"
"Invoke"
"BatchGetObject"
"GetDownloadUrlForLayer"
"GetObject"
"GetObject"
"CreateLogStream"
"AssumeRole"
"CreateLogStream"
"CreateLogStream"
"ListImages"
"CreateLogStream"
```

```
(shlomi@shlomi)-[~/../us-east-1/2018/11/28]
$ find . -type f -exec cat {} \; | jq -cr '.Records[]|.eventTime, .eventName|@tsv' | sort

2018-11-28T22:31:59Z      AssumeRole
2018-11-28T22:31:59Z      AssumeRole
2018-11-28T23:02:56Z      GetObject
2018-11-28T23:02:56Z      GetObject
2018-11-28T23:02:56Z      GetObject
2018-11-28T23:02:56Z      GetObject
2018-11-28T23:02:57Z      GetObject
2018-11-28T23:03:08Z      GetObject
2018-11-28T23:03:08Z      GetObject
2018-11-28T23:03:08Z      GetObject
2018-11-28T23:03:08Z      GetObject
2018-11-28T23:03:08Z      GetObject
2018-11-28T23:03:11Z      GetObject
2018-11-28T23:03:11Z      GetObject
2018-11-28T23:03:12Z      AssumeRole
2018-11-28T23:03:12Z      CreateLogStream
```

```
(shlomi@ ~) - ssh -i /home/shlomi/.ssh/sshkey.pem -o StrictHostKeyChecking=no -o UserKnownHostsFile=/dev/null -o LogLevel=ERROR -o ProxyCommand="ssh -W %h:%p -o StrictHostKeyChecking=no -o UserKnownHostsFile=/dev/null -o LogLevel=ERROR root@10.10.10.10" -o Port=22 root@10.10.10.10
$ find -type f -exec cat {} \; | jq -r '[_sourceIPAddress, .userIdentity.arn, .userIdentity.accountId, .userIdentity.type, .eventName]|@tsv' | sort

2018-11-28T22:31:59Z    ecs-tasks.amazonaws.com    AWSService    AssumeRole
2018-11-28T22:31:59Z    ecs-tasks.amazonaws.com    AWSService    AssumeRole
2018-11-28T23:02:56Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
2018-11-28T23:02:56Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
2018-11-28T23:02:56Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
2018-11-28T23:02:56Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
2018-11-28T23:02:57Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
2018-11-28T23:03:08Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
2018-11-28T23:03:08Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
2018-11-28T23:03:08Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
2018-11-28T23:03:08Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
2018-11-28T23:03:08Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
2018-11-28T23:03:11Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
2018-11-28T23:03:11Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
2018-11-28T23:03:12Z    34.234.236.212    arn:aws:sts::653711331788:assumed-role/level1/level1    653711331788    AssumedRole    CreateLogStream
2018-11-28T23:03:12Z    lambda.amazonaws.com    AWSService    AssumeRole
2018-11-28T23:03:13Z    34.234.236.212    arn:aws:sts::653711331788:assumed-role/level1/level1    653711331788    AssumedRole    CreateLogStream
2018-11-28T23:03:13Z    apigateway.amazonaws.com    AWSService    Invoke
2018-11-28T23:03:14Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
2018-11-28T23:03:17Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
2018-11-28T23:03:18Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
2018-11-28T23:03:20Z    34.234.236.212    arn:aws:sts::653711331788:assumed-role/level1/level1    653711331788    AssumedRole    CreateLogStream
2018-11-28T23:03:20Z    apigateway.amazonaws.com    AWSService    Invoke
2018-11-28T23:03:32Z    34.234.236.212    arn:aws:sts::653711331788:assumed-role/level1/level1    653711331788    AssumedRole    CreateLogStream
2018-11-28T23:03:50Z    34.234.236.212    arn:aws:sts::653711331788:assumed-role/level1/level1    653711331788    AssumedRole    CreateLogStream
2018-11-28T23:04:54Z    104.102.221.250    arn:aws:sts::653711331788:assumed-role/level1/level1    653711331788    AssumedRole    ListObjects
2018-11-28T23:05:10Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
2018-11-28T23:05:12Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
2018-11-28T23:05:12Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
2018-11-28T23:05:12Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
2018-11-28T23:05:53Z    104.102.221.250    arn:aws:sts::653711331788:assumed-role/level1/level1    653711331788    AssumedRole    ListImages
2018-11-28T23:06:17Z    104.102.221.250    arn:aws:sts::653711331788:assumed-role/level1/level1    653711331788    AssumedRole    BatchGetImage
2018-11-28T23:06:18Z    104.102.221.250    arn:aws:sts::653711331788:assumed-role/level1/level1    653711331788    AssumedRole    GetDownloadUrlForLayer
2018-11-28T23:07:08Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
2018-11-28T23:07:08Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
2018-11-28T23:09:28Z    104.102.221.250    arn:aws:sts::653711331788:assumed-role/level3/d190d14a-2404-45d6-9113-4eda22d7f2c7    653711331788    AssumedRole    ListBuckets
2018-11-28T23:09:36Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
2018-11-28T23:09:36Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
(shlomi@ ~) - ssh -i /home/shlomi/.ssh/sshkey.pem -o StrictHostKeyChecking=no -o UserKnownHostsFile=/dev/null -o LogLevel=ERROR -o ProxyCommand="ssh -W %h:%p -o StrictHostKeyChecking=no -o UserKnownHostsFile=/dev/null -o LogLevel=ERROR root@10.10.10.10" -o Port=22 root@10.10.10.10
```

```
n, .userIdentity.accountId, .userIdentity.type, .eventName]|@tsv' | sort > /home/shlomi/Desktop/report.txt
```

```
Text Editor
Simple Text Editor
~/Desktop/report.txt - Mousepad

1 2018-11-28T22:31:59Z    ecs-tasks.amazonaws.com    AWSService    AssumeRole
2 2018-11-28T22:31:59Z    ecs-tasks.amazonaws.com    AWSService    AssumeRole
3 2018-11-28T23:02:56Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
4 2018-11-28T23:02:56Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
5 2018-11-28T23:02:56Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
6 2018-11-28T23:02:56Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
7 2018-11-28T23:02:57Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
8 2018-11-28T23:03:08Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
9 2018-11-28T23:03:08Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
10 2018-11-28T23:03:08Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
11 2018-11-28T23:03:08Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
12 2018-11-28T23:03:08Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
13 2018-11-28T23:03:11Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
14 2018-11-28T23:03:11Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
15 2018-11-28T23:03:12Z    34.234.236.212    arn:aws:sts::653711331788:assumed-role/level1/level1    653711331788    AssumedRole    CreateLogStream
16 2018-11-28T23:03:12Z    lambda.amazonaws.com    AWSService    AssumeRole
17 2018-11-28T23:03:13Z    34.234.236.212    arn:aws:sts::653711331788:assumed-role/level1/level1    653711331788    AssumedRole    CreateLogStream
18 2018-11-28T23:03:13Z    apigateway.amazonaws.com    AWSService    Invoke
19 2018-11-28T23:03:14Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
20 2018-11-28T23:03:17Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
21 2018-11-28T23:03:18Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
22 2018-11-28T23:03:20Z    34.234.236.212    arn:aws:sts::653711331788:assumed-role/level1/level1    653711331788    AssumedRole    CreateLogStream
23 2018-11-28T23:03:20Z    apigateway.amazonaws.com    AWSService    Invoke
24 2018-11-28T23:03:32Z    34.234.236.212    arn:aws:sts::653711331788:assumed-role/level1/level1    653711331788    AssumedRole    CreateLogStream
25 2018-11-28T23:03:50Z    34.234.236.212    arn:aws:sts::653711331788:assumed-role/level1/level1    653711331788    AssumedRole    CreateLogStream
26 2018-11-28T23:04:54Z    104.102.221.250    arn:aws:sts::653711331788:assumed-role/level1/level1    653711331788    AssumedRole    ListObjects
27 2018-11-28T23:05:10Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
28 2018-11-28T23:05:12Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
29 2018-11-28T23:05:12Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
30 2018-11-28T23:05:53Z    104.102.221.250    arn:aws:sts::653711331788:assumed-role/level1/level1    653711331788    AssumedRole    ListImages
31 2018-11-28T23:06:17Z    104.102.221.250    arn:aws:sts::653711331788:assumed-role/level1/level1    653711331788    AssumedRole    BatchGetImage
32 2018-11-28T23:06:18Z    104.102.221.250    arn:aws:sts::653711331788:assumed-role/level1/level1    653711331788    AssumedRole    GetDownloadUrlForLayer
33 2018-11-28T23:07:08Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
34 2018-11-28T23:07:08Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
35 2018-11-28T23:09:28Z    104.102.221.250    arn:aws:sts::653711331788:assumed-role/level3/d190d14a-2404-45d6-9113-4eda22d7f2c7    653711331788    AssumedRole    ListBuckets
36 2018-11-28T23:09:36Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
37 2018-11-28T23:09:36Z    104.102.221.250    ANONYMOUS_PRINCIPAL    AWSAccount    GetObject
38
```

Objective 4 – Investigating Potential Credential Compromise

The final objective focused on identifying signs of a potential credential compromise by analyzing AWS CloudTrail logs and investigating suspicious API activity.

Identifying the ListBuckets API Call

The investigation began by filtering the CloudTrail logs for **ListBuckets** events using the following jq command:

```
```bash
jq '.Records[] | select(.eventName=="ListBuckets")'
```
```

Command Explanation

- **`.Records[]`** – Iterates through every CloudTrail event stored in the JSON document.
- **`.select()`** – Filters the results based on a specified condition.
- **`.eventName=="ListBuckets"`** – Returns only events where the executed AWS API operation was **ListBuckets**.

This query reduced the dataset to only bucket enumeration events, making it easier to identify which AWS identity performed the action.

The resulting output indicated that the **Level3** IAM role was responsible for executing the **ListBuckets** API request.

Investigating the IAM Role

To understand the permissions associated with the identified role, the following AWS CLI command was executed:

```
```bash
aws --profile target_security iam get-role --role-name Level3
```
```


'''

Command Explanation

- `--profile target_security` – Executes the request using the Target account profile.
- `iam` – Invokes the AWS Identity and Access Management service.
- `get-role` – Retrieves the configuration of an IAM role.
- `--role-name Level3` – Specifies the role to retrieve.

The command returned the complete IAM role configuration, including its trust policy and metadata.

The output showed that the `Level3` role was designed to be assumed by `Amazon ECS Tasks`, allowing workloads running inside Amazon ECS containers to securely interact with AWS services through temporary IAM credentials.

Security Analysis

Although the role was intended for Amazon ECS workloads, the CloudTrail logs revealed that a `ListBuckets` API request had been executed from the external IP address:

```text

104.102.221.250

'''

This IP address did not match the expected environment and therefore required additional investigation.

Because the API request originated from an unexpected source while using an ECS-related IAM role, the activity could indicate one of several security scenarios:

- Temporary credentials associated with the ECS task were exposed.
- The IAM role was assumed by an unauthorized principal.
- The credentials were used from an unexpected network location.
- Additional verification was required to determine whether the activity was legitimate or represented a credential compromise.

---

## ## Security Benefit

CloudTrail investigation combined with jq filtering enables security analysts to quickly identify suspicious AWS API activity, determine which IAM identity performed the action, and trace requests back to their originating IP address.

This methodology supports:

- Detection of compromised IAM credentials.
- Investigation of suspicious API calls.
- Validation of IAM role usage.
- Correlation between identities, permissions, and network activity.

- Faster incident response within AWS environments.

```
(shlomi@s)-[~/.../us-east-1/2018/11/28]
$ find . -type f -exec cat {} \; | jq '.Records[]|select(.eventName=="ListBuckets")'

{
 "eventVersion": "1.05",
 "userIdentity": {
 "type": "AssumedRole",
 "principalId": "AROAJQMBDNUMIKLZKMF64:d190d14a-2404-45d6-9113-4eda22d7f2c7",
 "arn": "arn:aws:sts::653711331788:assumed-role/level3/d190d14a-2404-45d6-9113-4eda22d7f2c7",
 "accountId": "653711331788",
 "accessKeyId": "ASIAZQNB3KHGNXWXBSJS",
 "sessionContext": {
 "attributes": {
 "mfaAuthenticated": "false",
 "creationDate": "2018-11-28T22:31:59Z"
 },
 "sessionIssuer": {
 "type": "Role",
 "principalId": "AROAJQMBDNUMIKLZKMF64",
 "arn": "arn:aws:iam::653711331788:role/level3",
 "accountId": "653711331788",
 "userName": "level3"
 }
 }
 },
 "eventTime": "2018-11-28T23:09:28Z",
 "eventSource": "s3.amazonaws.com",
 "eventName": "ListBuckets",
 "awsRegion": "us-east-1",
 "sourceIPAddress": "104.102.221.250",
 "userAgent": "[aws-cli/1.16.19 Python/2.7.10 Darwin/17.7.0 botocore/1.12.9]",
 "requestParameters": null,
 "responseElements": null,
 "requestID": "4698593B9338B27F",
 "eventID": "65e111a0-83ae-4ba8-9673-16291a804873",
 "eventType": "AwsApiCall",
 "recipientAccountId": "653711331788"
}

(shlomi@s)-[~/.../us-east-1/2018/11/28]
$
```

```

(shlomi@s)-[~/.../us-east-1/2018/11/28]
$ aws --profile target_security iam get-role --role-name level3
{
 "Role": {
 "Path": "/",
 "RoleName": "level3",
 "RoleId": "AROAJQMBDNUMIKLZKMF64",
 "Arn": "arn:aws:iam::653711331788:role/level3",
 "CreateDate": "2018-11-23T17:55:27+00:00",
 "AssumeRolePolicyDocument": {
 "Version": "2012-10-17",
 "Statement": [
 {
 "Sid": "",
 "Effect": "Allow",
 "Principal": {
 "Service": "ecs-tasks.amazonaws.com"
 },
 "Action": "sts:AssumeRole"
 }
]
 },
 "Description": "Allows ECS tasks to call AWS services on your behalf.",
 "MaxSessionDuration": 3600,
 "RoleLastUsed": {
 "LastUsedDate": "2025-04-14T01:11:39+00:00",
 "Region": "us-east-1"
 }
 }
}

(shlomi@s)-[~/.../us-east-1/2018/11/28]
$

```

shlomiGreen